

# Boundary and population effects under sharpening assignment policies

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September 17, 2026

## Abstract

Conversational AI systems choose persuasive strategies adaptively, and their logs are the evidence analysts inherit. We study what logs from a known, sharpening strategy-assignment policy reveal about the effect of the assigned strategy, with the text generator held fixed, and measure the cost of learning by cumulative exploration loss: the operational value forgone when the logged strategy differs from the one the deployer prefers. In a one-dimensional score model we prove a separation inside one outcome model. Under a single common temperature that shrinks with the sample size, the effect at the policy’s decision boundary admits asymptotically valid Wald inference while exploration loss tends to zero. In the same model, outcome laws with equal boundary effects but different population average effects become indistinguishable whenever exploration loss tends to zero, and uniformly consistent estimation of the population effect requires divergent worst-case loss. Without extrapolation structure, an information–exploration inequality for all estimators and adaptive designs shows that uniform precision  $\delta$  requires worst-case loss of order at least  $(\mathbb{E}\sqrt{g})^2/\delta^2$ , independent of  $n$ . For small fixed  $\delta$ , one deterministic common temperature requires loss of order at least  $n/\log^2(n\delta^2)$ , whereas uniform mixing keeps it bounded.

**Short title:** When the treatment talks back

**Keywords:** adaptive experiments, limited overlap, regression discontinuity, exploration, generative AI, off-policy evaluation

**MSC 2020:** 62D20, 62K05, 62C20, 62G20

## 1 Introduction

Conversational artificial intelligence persuades. In randomized trials, GPT-4 shifted debate opponents’ positions more often than human debaters did [1], and conversations with

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frontier models moved policy attitudes with effects that persisted for weeks [2]. Establishing that a technology persuades is the first question, not the last. Regulation and replication concern narrower objects: whether presenting evidence rather than asking a question, at this point in this conversation, changes what a person believes. Systems that persuade at scale choose such strategies adaptively, and their logs are what an analyst inherits. One of the strongest levers identified by Hackenburg et al. [2], a reward model that selects among candidate responses, is itself an adaptive selection policy.

This paper asks a precise version of the question those logs raise. Suppose the deployer’s strategy policy is known, as it is when the deployer records action probabilities, and suppose it sharpens as the deployer becomes confident, for instance by lowering a softmax temperature. Which causal effects can the logs estimate, and what must be paid to estimate the others?

The answer depends on the target. We distinguish two:

- the *boundary effect*, the effect among contexts where the policy is indifferent between strategies; and
- the *population average effect* over all contexts the deployment serves.

These answer different questions. The first is relevant to marginal changes of a policy that is already deployed; the second to whether a strategy should be used at all. Estimating one well does not substitute for estimating the other.

We price learning in the deployer’s own currency. Each time the logged strategy differs from the one the deployer’s objective prefers, the deployer gives up an operational value gap  $g(h)$ . The cumulative sum is the *exploration loss*  $R_n$ . This quantity is defined by the deployer’s objective, not by the human outcome under study, so it can be computed at the design stage without knowing the effects being estimated.

## Contributions.

1. *Separation in one model* (Theorem 3). In a one-dimensional score model with Lipschitz mean outcomes near the decision boundary, logs generated by a single common temperature  $\tau_n = n^{-\zeta}$ ,  $\zeta \in (1/2, 1)$ , give asymptotically valid Wald intervals for the boundary effect at every member of the model, while  $R_n \rightarrow 0$  (Theorem 1). In the same model, whenever  $R_n \rightarrow 0$ , two members that coincide near the boundary but differ in the population effect have total variation distance tending to zero, so no estimator is consistent for the population effect at both; and uniform consistency for the population effect requires  $\sup R_n \rightarrow \infty$ . Figure 1 shows the separation in exact risks for a simple design.
2. *An information–exploration inequality* (Theorem 2). In the model of all bounded mean functions, for every estimator and every adaptive or non-adaptive design, the product of worst-case mean squared error and worst-case exploration loss plus a constant is at least  $\sigma^2(\sum_k f_k \sqrt{g_k})^2$ . Uniform precision  $\delta$  therefore requires worst-case exploration loss at least of order  $(\mathbb{E}\sqrt{g})^2/\delta^2$ , independent of  $n$ ; a design exploring with probability proportional to  $g^{-1/2}$  attains this order up to constants (Proposition 1).

3. *What this means for temperature.* For any sufficiently small fixed precision  $\delta$ , a single deterministic common temperature requires exploration loss at least of order  $n/\log^2(n\delta^2)$  (Theorem 4), against a loss of order  $\delta^{-2}$ , bounded in  $n$ , for the designs below. Context-dependent temperatures implement the gap-based design (Proposition 2). Random mixtures of finite temperatures can also avoid the rate (Remark 1). When the value gap is misspecified, a Kantorovich bound says when gap-based allocation is worth its risk relative to uniform mixing (Lemma 1).

**Scope.** All results are for one decision per unit with independent contexts and a known assignment policy. Theorem 1 and Theorem 3 use a one-dimensional score, and their boundary guarantees are pointwise in the outcome law. The impossibility results hold in models without extrapolation structure: where data near the boundary identify the population effect, as under a constant effect, the separation disappears. Multi-turn conversations, estimated policies, and effects of realized text features are outside the formal results.

**Relation to existing work.** *Boundary estimands from algorithmic assignment.* Narita and Yata [3] show that weights of overlap type, built from an approximate propensity score of a known algorithm, converge to a boundary estimand and require undersmoothing; this is the closest antecedent for Theorem 1. There the smoothing is an analyst-chosen ball in covariate space; here it is a temperature chosen by the system in score space, and we compute the exploration loss jointly with the inferential guarantee. Eckles et al. [4] use known assignment noise near a cutoff for design-based inference, with a fixed noise scale and no cost accounting. Overlap weights and their target population are due to Li, Morgan and Zaslavsky [5]; they note that the weights peak where the propensity score is one half, but do not study a sharpening limit.

*Exploration and estimation.* Simchi-Levi and Wang [6] show a regret–estimation trade-off in multi-armed bandits without contexts; Wan, Kveton and Song [7] optimize exploration for policy evaluation under a value constraint; Douglas, Persson and Provost [8] derive optimal logging policies for inverse-propensity estimation and report that softmax logging performs worse than top- $k$  logging in simulation; Koppel et al. [9] extend inverse-gap-weighting contextual bandits to balance regret against the quality of average-treatment-effect estimates (we have read only the abstract of that workshop paper). The allocation  $q \propto g^{-1/2}$  has the form of optimum stratified allocation with costs [10]; we do not claim the allocation as new. Cesa-Bianchi et al. [11] show that Boltzmann exploration with a common schedule can fail for regret and that arm-dependent schedules repair it; our Theorem 4 and Proposition 2 are estimation analogues.

*Limited overlap and off-policy evaluation.* Lower bounds for off-policy value estimation [12–14] and limited-overlap asymptotics [15–17] supply the information-theoretic ingredients. Khan and Tamer [15] observe that the average treatment effect loses identification when any region of the covariate support is removed, and derive rates for trimmed inverse weighting when propensities approach zero or one in the covariate tails; their propensities do not change with  $n$ , and there is no cost of exploration. What we add is the coupling

of the information bound to the exploration loss, and the comparison of boundary and population targets on that common scale. Khan, Saveski and Ugander [18] bound effects without overlap under smoothness, which is the kind of extrapolation structure our impossibility excludes.

*Generative treatments.* Athey, Imbens and Ji [19] identify effects of content features from stochastic algorithms using logged unexposed candidates and probability ratios. Their estimand averages over co-occurring pairs with trimming and does not conflict with our bounds, which concern the population effect of a strategy under outcomes observed only for the logged action. Fresh and Shin [20] give a taxonomy of conversational estimands; we work at the level of strategy assignment. Imai and Nakamura [21] and Nakamura and Imai [22] study effects of realized text features under deterministic decoding; we instead hold the generation kernel fixed.

## 2 Setup

**Generative treatments.** Units  $i = 1, \dots, n$  arrive with contexts  $H_i \sim F$ , independently. The system chooses a strategy  $A_i \in \{0, 1\}$  and produces a message  $M_i \sim G(\cdot \mid A_i, H_i)$  from a fixed generation kernel  $G$ , which includes token-level sampling settings. The outcome is  $Y_i$ . The treatment is assignment of the strategy with  $G$  held fixed: potential outcomes  $Y_i(a)$  are defined under the kernel intervention that sets  $A_i = a$  and draws the message from  $G$  [23, 24]. A label attached to a realized message after the fact is a different object, and changing  $G$ , for example by changing token temperature, changes the treatment. We assume:

- (C1) *consistency*:  $Y_i = Y_i(A_i)$ , where the randomness of the message drawn from  $G$  is part of  $Y_i(a)$ ;
- (C2) *independent units*:  $(H_i, Y_i(0), Y_i(1))$  are independent and identically distributed across  $i$ , with no interference;
- (C3) *known sequential assignment*: given  $H_i$  and the past data  $\mathcal{F}_{i-1}$ ,  $A_i$  is drawn with known probabilities, independently of  $(Y_i(0), Y_i(1))$  and of all future units.

Under (C1)–(C3),  $\mathbb{E}[Y_i \mid A_i = a, H_i = h, \mathcal{F}_{i-1}] = \mathbb{E}[Y(a) \mid H = h]$ , and the law of  $Y_i$  given  $(A_i = a, H_i = h, \mathcal{F}_{i-1})$  is that of  $Y(a)$  given  $H = h$ . All statements below about conditional outcome laws are therefore statements about potential outcomes.

**Deployer.** The deployer has a score gap  $\Delta(h) = s_1(h) - s_0(h)$  from its own objective, a greedy strategy  $a^*(h) = \mathbf{1}\{\Delta(h) > 0\}$ , and an operational value gap  $g(h) \geq 0$ , the value its objective loses by choosing  $\bar{a}(h) = 1 - a^*(h)$ . We write  $g = \phi(|\Delta|)$  when the gap is a function of the score.

**Designs and notation for probabilities.** We use two probabilities and keep them distinct:

$$e(h) = \mathbb{P}(A = 1 \mid H = h), \quad q(h) = \mathbb{P}(A \neq a^*(h) \mid H = h),$$

the probability of strategy 1 and the probability of the off-greedy strategy. A design specifies  $Q_i = \mathbb{P}(A_i = \bar{a}(H_i) \mid H_i, \mathcal{F}_{i-1})$ ; it is non-adaptive if  $Q_i = q(H_i)$ . A *common temperature* design uses one deterministic  $\tau = \tau_n$  and

$$e_\tau(h) = \text{sig}(\Delta(h)/\tau), \quad q_\tau(h) = \text{sig}(-|\Delta(h)|/\tau),$$

where  $\text{sig}(x) = 1/(1 + \exp(-x))$ .

**Exploration loss.**  $R_n = \mathbb{E}[\sum_{i=1}^n g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}]$ ; for non-adaptive designs  $R_n = n\mathbb{E}[g(H)q(H)]$ .

**Estimands.** Let  $\mu_a(h) = \mathbb{E}[Y(a) \mid H = h]$  and  $c = \mu_1 - \mu_0$ . The population average effect is  $\theta = \mathbb{E}[c(H)]$ . With a one-dimensional score  $\Delta(h) = h$ , the boundary effect is  $\beta_0 = c(0)$ . For a given temperature, the overlap-weighted effect  $\beta_{\text{ov},\tau} = \mathbb{E}[\omega_\tau(H)c(H)]/\mathbb{E}[\omega_\tau(H)]$  with  $\omega_\tau = e_\tau(1 - e_\tau)$  [5] is a third, temperature-indexed, target; we use it only as an intermediate object.

### 3 Boundary effects at vanishing exploration loss

**Assumption 1** (Boundary model).  $\Delta(h) = h$ ,  $a^*(h) = \mathbf{1}\{h > 0\}$ , and

- (D1)  $H$  has a density  $f \leq \bar{f}$ , continuous at 0, with  $f(0) > 0$ ;
- (D2)  $|\mu_a| \leq B$ , and  $\mu_a$  is  $L$ -Lipschitz on  $[-u_0, u_0]$ ;
- (D3)  $\sigma_a^2(h) = \text{Var}(Y \mid A = a, H = h)$  is continuous at 0 with  $\sigma_1^2(0) + \sigma_0^2(0) > 0$ , and  $\mathbb{E}[Y^4 \mid A, H] \leq M$ ;
- (D4)  $\mathbb{P}(A = 1 \mid H = h) = \text{sig}(h/\tau_n)$  with  $\tau_n \rightarrow 0$ , known;
- (D5)  $g = \phi(|h|) \leq \bar{G}$  with  $\phi(u) \leq \bar{\kappa}u$  on  $[0, u_0]$ ; in the strong form  $\phi'(0) = \kappa > 0$ .

With  $e = e_{\tau_n}$ ,  $w_1 = A(1 - e)$  and  $w_0 = (1 - A)e$ , let  $\hat{\beta} = \sum w_{1i}Y_i / \sum w_{1i} - \sum w_{0i}Y_i / \sum w_{0i}$ , with variance estimator  $\hat{s}^2 = n^{-1} \sum \hat{\psi}_i^2$ , where  $\hat{\psi} = w_1(Y - \hat{m}_1)/\hat{D}_1 - w_0(Y - \hat{m}_0)/\hat{D}_0$ ,  $\hat{D}_a = n^{-1} \sum w_{ai}$  and  $\hat{m}_a$  is the weighted mean in arm  $a$ .

**Theorem 1** (Boundary inference). *Under Assumption 1, if  $n\tau_n \rightarrow \infty$ :*

- (i)  $(\hat{\beta} - \beta_{\text{ov},\tau_n})/(s_n/\sqrt{n}) \Rightarrow N(0, 1)$  with  $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau_n f(0)\}(1 + o(1))$ ;
- (ii)  $|\beta_{\text{ov},\tau_n} - \beta_0| \leq C\tau_n$  for large  $n$ ;
- (iii)  $\hat{s}^2/s_n^2 \rightarrow 1$  in probability;
- (iv) if also  $n\tau_n^3 \rightarrow 0$ , the Wald interval  $\hat{\beta} \pm z_{1-\eta/2}\hat{s}/\sqrt{n}$  has asymptotic coverage  $1 - \eta$  for  $\beta_0$ ;
- (v)  $R_n = n\mathbb{E}[g(H)q_{\tau_n}(H)] \leq n\tau_n^2\bar{\kappa}\bar{f}\pi^2/6 + n\bar{G}\exp(-u_0/\tau_n)$ , and in the strong form  $R_n = n\tau_n^2\kappa f(0)\pi^2/6(1 + o(1))$ .

In particular, for  $\tau_n = n^{-\zeta}$  with  $\zeta \in (1/2, 1)$ , the interval in (iv) is asymptotically valid and  $R_n \rightarrow 0$ .

The three rate conditions are  $n\tau_n \rightarrow \infty$  (the boundary sample grows,  $\zeta < 1$ ),  $n\tau_n^3 \rightarrow 0$  (bias is negligible relative to the standard error,  $\zeta > 1/3$ ), and  $n\tau_n^2 \rightarrow 0$  (the loss vanishes,  $\zeta > 1/2$ ). The error  $|\hat{\beta} - \beta_0| = O_p((n\tau_n)^{-1/2})$  is achievable; we do not claim it is optimal, and it is a statement in probability rather than in mean square. Differentiability of  $c$  at zero does not improve the bias order in general:  $c(h) = 1 + |h|^{3/2}$  gives bias proportional to  $\tau^{3/2}$ . If  $c$  is  $C^{1,1}$  and  $f$  is Lipschitz near zero, the bias is  $O(\tau^2)$  and condition (iv) relaxes to  $n\tau_n^5 \rightarrow 0$ . Proofs are in Appendix A.

Vanishing loss does not mean that exploration disappears. The expected number of off-greedy actions is  $n\mathbb{E}[q_{\tau_n}(H)] = 2 \log 2 f(0) n\tau_n(1 + o(1))$ , which grows because  $n\tau_n \rightarrow \infty$ ; the loss nevertheless vanishes,  $R_n \asymp n\tau_n^2$ , because these actions occur ever closer to the boundary, where the value gap is small.

## 4 Population effects require paid exploration

Let  $\mathcal{M}(B, \sigma)$  be the model in which  $Y \mid A = a, H = h$  is  $N(\mu_a(h), \sigma^2)$  and  $\mu_0, \mu_1$  are any measurable functions bounded by  $B$ . The model contains no structure that carries effects learned near the boundary to other contexts.

**Theorem 2** (Information–exploration inequality). *Let  $S_1, \dots, S_K$  be disjoint sets with  $f_k = F(S_k) > 0$ , each contained in  $\{a^* = 0\}$  or in  $\{a^* = 1\}$ , and  $g_k = \inf_{S_k} g$ . For every design, adaptive or not, and every estimator  $\hat{\theta}$ ,*

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \cdot \left( \sup_{\mathcal{M}(B, \sigma)} R_n + \frac{\sigma^2 \pi^2}{B^2} \sum_k g_k \right) \geq \sigma^2 \left( \sum_k f_k \sqrt{g_k} \right)^2. \quad (1)$$

Consequently, if  $F(g \geq \gamma) > 0$  for some  $\gamma > 0$ , uniform consistency requires  $\sup R_n \rightarrow \infty$ ; and if  $\sup \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2 \rightarrow 0$ , then  $\liminf_n \delta_n^2 \sup R_n \geq \sigma^2 (\mathbb{E} \sqrt{g(H)})^2$ .

Neither side of (1) involves  $n$ : deployment scale does not substitute for exploration when the target is the population effect. At  $R_n = 0$  the bound is the finite floor  $B^2 (\sum f_k \sqrt{g_k})^2 / (\pi^2 \sum g_k)$ , consistent with the fact that a constant estimator has bounded error.

**Proposition 1** (Achievability). *Suppose  $\mathbb{E} \sqrt{g} > 0$  and  $n\delta_n^2 \rightarrow \infty$ . The non-adaptive design  $q(h) = \min\{1/2, c_n / \sqrt{g(h)}\}$  with  $c_n = \mathbb{E} \sqrt{g} / \{n\delta_n^2 / (\sigma^2 + B^2) - 4\}$  and the Horvitz–Thompson estimator give  $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$  and  $R_n \leq (\sigma^2 + B^2) (\mathbb{E} \sqrt{g})^2 \delta_n^{-2} (1 + o(1))$ . With  $C = \sigma^2 + B^2$  and  $n\delta_n^2 \geq 4C$ , uniform mixing  $q \equiv C / (n\delta_n^2 - 2C)$  also gives  $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$ , with  $R_n = C \mathbb{E}[g] \delta_n^{-2} (1 + o(1))$ .*

The order  $\delta^{-2}$  and the functional  $(\mathbb{E} \sqrt{g})^2$  therefore appear in both the lower bound and the attainable loss. The constants differ: the lower bound has  $\sigma^2$ , and the Horvitz–Thompson upper bound retains  $B^2$ . We do not establish the optimal constant. Proofs are in Appendix B.

## 5 The separation

Theorem 2 is proved in  $\mathcal{M}(B, \sigma)$ , while Theorem 1 assumes Lipschitz means near the boundary. A lower bound over a large model does not transfer to a smaller one automatically. We therefore state both sides in one model.

Let  $\mathcal{M}_D$  be the set of outcome laws satisfying (D1), (D2), Gaussian noise with variance  $\sigma^2$ , and (D5) with  $\phi$  not necessarily monotone. Under Gaussian noise, (D3) holds for every member of  $\mathcal{M}_D$ .

**Assumption 2.** There are  $u_0 < u_1 < u_2$  with  $F([u_1, u_2]) > 0$  and  $\gamma := \text{ess inf}_{h \in [u_1, u_2]} g(h) > 0$ , and  $B, L > 0$ .

**Theorem 3** (Separation). *Under Assumption 2, let  $b(h) = \sin^2(\pi(h-u_1)/(u_2-u_1))\mathbf{1}\{u_1 \leq h \leq u_2\}$ ,  $t_{\max} = \min\{B, L(u_2-u_1)/\pi\}$ , and for  $|t| \leq t_{\max}$  let  $\mu^t \in \mathcal{M}_D$  have  $\mu_1^t \equiv 0$  and  $\mu_0^t = -tb$ . All  $\mu^t$  coincide on  $(-\infty, u_1]$ , have boundary effect  $\beta_0 = 0$ , and have population effect  $\theta(\mu^t) = t\mathbb{E}[b(H)]$ .*

(S1) *Under the common-temperature design  $\tau_n = n^{-\zeta}$ ,  $\zeta \in (1/2, 1)$ ,  $R_n \rightarrow 0$  and, for every  $\mu \in \mathcal{M}_D$ , the Wald interval of Theorem 1 has asymptotic coverage  $1 - \eta$  for  $\beta_0(\mu)$ .*

(S2) *For every design and every  $t$ ,  $\text{TV}(P_{\mu^0}^n, P_{\mu^t}^n) \leq R_n(\mu^0)/\gamma$ . Hence if  $R_n(\mu^0) \rightarrow 0$ , for every  $t_1 \in (0, t_{\max}]$  and every estimator,  $\max_{t \in \{0, t_1\}} P_{\mu^t}(|\hat{\theta}_n - \theta(\mu^t)| \geq t_1 \mathbb{E}[b]/2) \geq (1 - R_n(\mu^0)/\gamma)/2 \rightarrow 1/2$ .*

(S3) *For every design and every estimator,*

$$\sup_{\mathcal{M}_D} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \frac{(\mathbb{E}[b])^2}{\sup_{\mathcal{M}_D} R_n/(\sigma^2\gamma) + \pi^2/t_{\max}^2}.$$

Under the design of (S1),  $\mu^0$  and  $\mu^{t_1}$  are both members of  $\mathcal{M}_D$  with the same boundary effect, and the boundary interval is valid at each. By (S2), no estimator of the population effect is consistent at both. By (S3), consistency uniformly over  $\mathcal{M}_D$  requires unbounded exploration loss. We do not claim that the constant of Theorem 2 holds in  $\mathcal{M}_D$ , nor any uniform guarantee on the boundary side. If a submodel lets boundary data identify  $\theta$ , the perturbation family is excluded and (S2)–(S3) do not apply. Proofs are in Appendix C.

Figure 1 illustrates the separation with exact risks, computed by numerical integration, for the design used in Section 7. Under one common temperature, the population effect reaches root mean squared error 0.1 only where the exploration loss is large: on the plotted grid the smallest such loss is about 102. The boundary effect reaches the same precision with loss 0.58 at the edge of the grid, and with less loss at larger  $n$  and sharper policies.

## 6 Temperature as an exploration device

**Theorem 4** (One common temperature). *Consider common-temperature designs with one deterministic  $\tau_n$  per  $n$ . Assume  $|\Delta(H)|$  has a density at least  $f_{\min} > 0$  on  $[0, u_0]$ ;  $\phi(u) \geq \kappa u$*

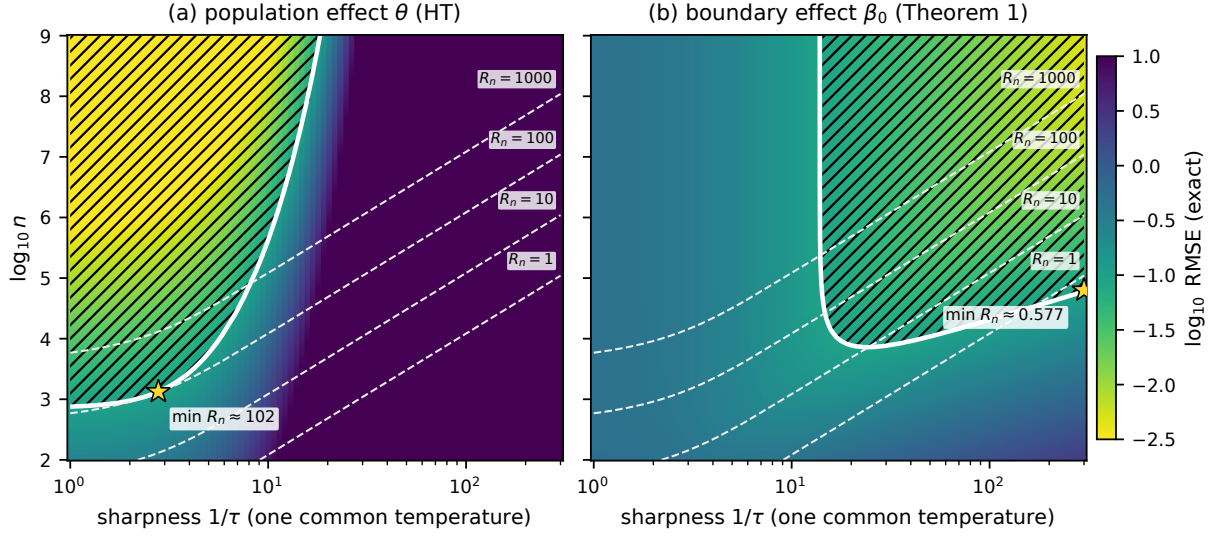


Figure 1: Exact root mean squared error (colour,  $\log_{10}$  scale) under one common temperature  $\tau$  for the design of Section 7:  $H \sim U(-1, 1)$ ,  $\Delta(h) = h$ ,  $g(h) = |h|$ ,  $c(h) = 1 + |h|$ , unit noise. (a) The Horvitz–Thompson estimator of the population effect  $\theta$ . (b) The estimator of Theorem 1 for the boundary effect  $\beta_0$ , including its bias. Hatched regions, bounded by white curves, have root mean squared error at most 0.1. Dashed curves are levels of the cumulative exploration loss  $R_n = n\mathbb{E}[g(H)q_\tau(H)]$ . Stars mark the smallest loss that attains the precision on the plotted grid ( $1 \leq 1/\tau \leq 300$ ,  $10^2 \leq n \leq 10^9$ ). Panel (a) concerns one estimator under one outcome law; the lower bounds for all estimators are Theorems 2 and 3.

on  $[0, u_0]$ ;  $G = \text{ess sup } |\Delta(H)| < \infty$ ; and for every  $\eta' \in (0, G)$ ,  $\nu_{\eta'} = \max_j \mathbb{P}(|\Delta(H)| \geq G - \eta', a^*(H) = j) > 0$ . Fix  $\eta' \in (0, G)$ . If an estimator has  $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$  with  $\delta_n^2 \leq \nu_{\eta'}^2 B^2 / (2\pi^2)$  and  $2n\delta_n^2 > \nu_{\eta'} \sigma^2$ , then

$$R_n \geq \kappa f_{\min} c_0 n \min \left\{ u_0^2, \frac{(G - \eta')^2}{\log^2 \{2n\delta_n^2 / (\nu_{\eta'} \sigma^2)\}} \right\}, \quad c_0 = \int_0^1 v \text{sig}(-v) dv.$$

This is a lower bound, and  $\delta_n$  may be fixed. For any fixed  $\delta$  with  $\delta^2 \leq \nu_{\eta'}^2 B^2 / (2\pi^2)$ , the common-temperature loss is at least of order  $n / \log^2 n$ , whereas Proposition 1 attains the same uniform precision with loss bounded in  $n$ . The precision condition cannot be dropped: for  $\delta \geq 2B$  the estimator that always outputs zero meets the target with no exploration. Relative to the attainable loss, the lower bound is larger by a factor of order at least  $n\delta_n^2 / \log^2(n\delta_n^2)$ , which diverges whenever  $n\delta_n^2 \rightarrow \infty$ . With finitely many score values  $0 < \Delta_{\min} < \Delta_{\max}$  and fixed, sufficiently small  $\delta$ , the common-temperature loss is at least of order  $n^{1-\Delta_{\min}/\Delta_{\max}}$ . The reason is not that a common temperature explores in the wrong place: the optimal design also explores most near the boundary, where exploration is cheap. It is that the exploration probability of a common temperature decays exponentially in  $|\Delta|$ , so observations far from the boundary become too rare unless the temperature is raised everywhere, which is expensive near the boundary.

*Remark 1* (Mixtures of temperatures). Theorem 4 does not extend to random mixtures of finite temperatures. With  $|\Delta| = |H|$ ,  $H \sim U(-1, 1)$ ,  $g = |H|$ , mixing temperature 1 with weight  $w_n = \mathbb{E}[1/\text{sig}(-|H|)]/(n\delta^2)$  and temperature  $1/n$  otherwise keeps the design variance  $\mathbb{E}[1/q]/n$  at most  $\delta^2$ , for fixed  $\delta$ , with loss bounded in  $n$  (46.36 at  $\delta = 0.1$  for  $n = 10^4, 10^5, 10^6$ ). The lesson concerns a single common temperature, not temperature as such.

**Proposition 2** (Context temperatures). *For  $\Delta(h) \neq 0$  and  $q(h) \in (0, 1/2)$ , the context temperature  $\tau(h) = |\Delta(h)| / \log\{(1 - q(h))/q(h)\}$  gives off-greedy probability  $q(h)$ . The value  $q(h) = 1/2$  requires  $\tau(h) = \infty$ , and on  $\{\Delta = 0\}$  every temperature gives  $q = 1/2$ . Suppose  $g = \phi(|\Delta|)$  with  $\phi(0) = 0$ , so that  $g = 0$  on  $\{\Delta = 0\}$ . Then the design  $q^*$  of Proposition 1 is implementable with temperatures in  $(0, \infty]$ :  $q^* = 1/2$  wherever  $g = 0$ , and for each  $h$  with  $\Delta(h) \neq 0$  and  $g(h) > 0$ ,  $q^*(h) < 1/2$  for large  $n$  and  $\tau^*(h) \log(1/c_n) \rightarrow |\Delta(h)|$ .*

The limit is pointwise for fixed  $h$  with  $g(h) > 0$ . It does not say that the implementing temperature is small near the boundary at finite  $n$ : where the cap binds, for example  $0 < |h| \leq 4c_n^2$  when  $g(h) = |h|$ , the implementing temperature is  $\tau = \infty$ .

**Lemma 1** (Misspecified gaps). *Compare the designs  $q = c/\sqrt{\hat{g}}$  and  $q = c'/\sqrt{g}$ , with  $c, c'$  chosen so that both have the same design variance  $\sigma^2 \mathbb{E}[1/q]/n$ , and assume the cap is slack, that is, both designs are at most  $1/2$  on  $\{g > 0\}$ . Let  $\hat{g}/g \in [a, b]$  on  $\{g > 0\}$  and  $\rho = b/a$ . Then the loss ratio is at most  $K(\rho) = (1 + \sqrt{\rho})^2 / (4\sqrt{\rho})$ . The bound is attained when the measure  $w \propto \sqrt{g} dF$  has a set of mass exactly one half, for example when it is non-atomic.*

A common rescaling of  $\hat{g}$  changes nothing; only relative misspecification across contexts matters. Gap-based allocation beats uniform mixing for every admissible  $\hat{g}$  if  $\mathbb{E}[g]/(\mathbb{E}\sqrt{g})^2 > K(\rho)$ ; the condition is also necessary when the bound is attained, but not in general. When the cap binds, the ratio bound holds in the limit of large design variance. Suppose  $0 < a \leq \hat{g}/g \leq b < \infty$  on  $\{g > 0\}$ ,  $\hat{g} = 0$  on  $\{g = 0\}$ ,  $0 < \mathbb{E}\sqrt{g} < \infty$ , and let  $q_c = \min\{1/2, c/\sqrt{\hat{g}}\}$  with  $q_c = 1/2$  where  $\hat{g} = 0$ . By dominated convergence,  $c\mathbb{E}[1/q_c] = \mathbb{E}\max\{2c, \sqrt{\hat{g}}\} \rightarrow \mathbb{E}\sqrt{\hat{g}}$  and  $\mathbb{E}[gq_c]/c \rightarrow \mathbb{E}[g/\sqrt{\hat{g}}]$  (with  $g/\sqrt{\hat{g}} = 0$  where  $g = 0$ ) as  $c \rightarrow 0$ . Calibrating  $c$  to a common design variance  $T = \mathbb{E}[1/q_c] \rightarrow \infty$ , the loss ratio converges to  $\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/(\mathbb{E}\sqrt{g})^2$ , the quantity bounded in the lemma. Proofs are in Appendix D.

## 7 Simulations

All simulations use  $H \sim U(-1, 1)$ ,  $\Delta(h) = h$ ,  $g(h) = |h|$ ,  $\mu_0(h) = h$ ,  $c(h) = 1 + |h|$  (population effect 1.5, boundary effect 1), and  $N(0, 1)$  noise.

**Exploration designs.** At  $n = 10^5$  and design variance  $\delta^2 = 0.01$ , the closed-form exploration loss is 50.0 for uniform mixing, 44.4 for  $q \propto g^{-1/2}$ , and 988.2 for a common temperature ( $\tau = 0.110$ ), a ratio of 19.8 to uniform mixing. Table 1 runs the augmented inverse-propensity estimator with a linear working model on common random numbers across designs. The context-temperature implementation of  $q \propto g^{-1/2}$  produces identical rows and is not listed separately. The designs are matched on design variance, not on realized RMSE. The working model is correctly specified for this data-generating process, so the table illustrates the designs rather than verifying the uniform Horvitz–Thompson guarantee of Proposition 1. At a fixed  $\delta$ , the expected number of off-greedy actions does not grow with  $n$ , so these Wald intervals need not approach nominal coverage as  $n$  grows. In a two-context example the bound (1) evaluates to 51.9 against the asymptotic constant 52.4, and the common-temperature loss has local growth exponent 0.808 against the predicted 0.8.

| $\delta$ | design                                 | RMSE  | coverage | MCSE  | explored | loss $R_n$ |
|----------|----------------------------------------|-------|----------|-------|----------|------------|
| 0.10     | uniform mixing                         | 0.098 | 0.950    | 0.011 | 100      | 49.9       |
| 0.10     | $q \propto g^{-1/2}$ (= context temp.) | 0.103 | 0.932    | 0.013 | 134      | 44.5       |
| 0.10     | common temperature                     | 0.097 | 0.965    | 0.009 | 7607     | 988.5      |
| 0.05     | uniform mixing                         | 0.052 | 0.955    | 0.010 | 400      | 200.3      |
| 0.05     | $q \propto g^{-1/2}$ (= context temp.) | 0.053 | 0.940    | 0.012 | 532      | 178.0      |
| 0.05     | common temperature                     | 0.050 | 0.948    | 0.011 | 9191     | 1440.1     |

Table 1: Exploration designs at  $n = 10^5$ , 400 replications with common random numbers. Coverage is of the population effect by Wald intervals; MCSE uses the observed coverage. “Explored” is the mean number of off-greedy actions.

**Boundary inference.** Table 2 reports the boundary estimator of Theorem 1. Exploration loss matches the constant in Theorem 1(v). With  $\zeta = 0.6$  coverage is close to nominal; at  $n = 10^6$ , 5,000 replications give 0.9458 (MCSE 0.0032), with the estimated standard error at 0.993 of the Monte Carlo standard deviation. With a very small boundary sample the interval under-covers: 0.887 at  $n\tau_n \approx 10$ . The guarantee of Theorem 1 is asymptotic.

**Temperature sweep.** Figure 2 fixes  $n = 100,000$  and sharpens a common temperature, with 1,000 replications per temperature. Wald coverage of the population effect falls from 0.949 at  $1/\tau = 1$  to 0.085 (Horvitz–Thompson) and 0.031 (augmented estimator) at  $1/\tau = 120$ . Over the same range, coverage of the boundary effect rises from 0.000 to 0.943, and the exploration loss falls from 17,056 to 5.7. The Monte Carlo error of the population estimators understates their exact standard deviation once the policy is sharp: at  $1/\tau = 120$  the Monte Carlo root mean squared error of the Horvitz–Thompson estimator is 0.36, while its exact standard deviation is of order  $10^{22}$ . Its variance is carried by rare, very large terms that the replications do not observe. For the population estimators, the exact curves rather than the Monte Carlo points describe the error.

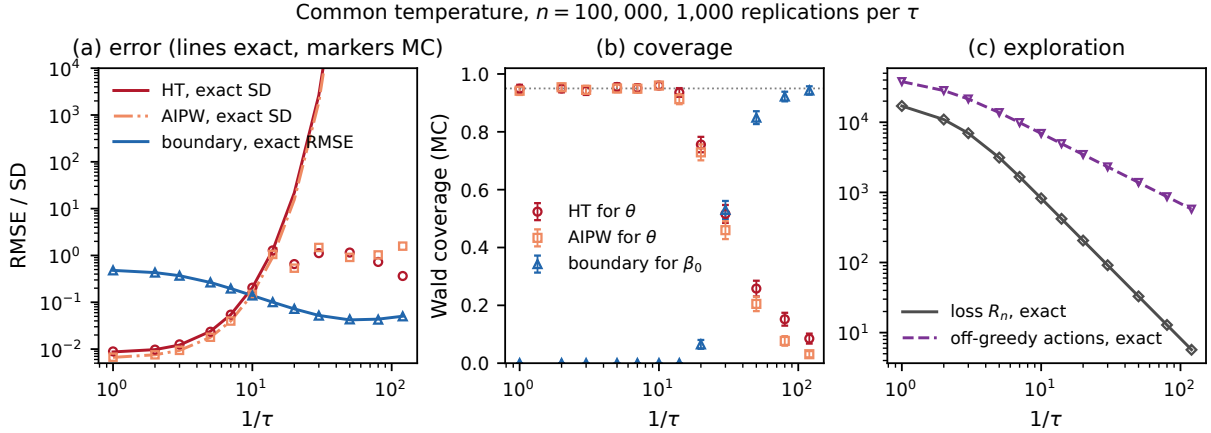


Figure 2: Common temperature at fixed  $n = 100,000$ , 1,000 replications per temperature, same design as Figure 1. (a) Lines: exact standard deviation of the Horvitz–Thompson and augmented (true regression) estimators of  $\theta$ , and exact root mean squared error of the boundary estimator of  $\beta_0$ . Markers: Monte Carlo root mean squared error; the augmented estimator uses a linear working model. (b) Wald coverage with 95% Monte Carlo intervals. Boundary coverage is low while  $n\tau^3$  is large, as Theorem 1(iv) requires. (c) Exploration loss  $R_n$  and expected number of off-greedy actions; lines are exact, markers Monte Carlo.

## 8 Discussion

The results separate two uses of adaptive logs. If the question is how outcomes respond where the deployed policy is indifferent, logs from a sharpening policy are informative, and

| $\zeta$ | $n$       | $n\tau_n$ | reps  | RMSE  | coverage of $\beta_0$ | MCSE  | loss $R_n$ (Thm. 1)                           |
|---------|-----------|-----------|-------|-------|-----------------------|-------|-----------------------------------------------|
| 0.60    | 10,000    | 39.8      | 1,000 | 0.215 | 0.949                 | 0.007 | $1.3 \times 10^{-1}$ ( $1.3 \times 10^{-1}$ ) |
| 0.60    | 100,000   | 100.0     | 500   | 0.141 | 0.944                 | 0.010 | $8.2 \times 10^{-2}$ ( $8.2 \times 10^{-2}$ ) |
| 0.60    | 1,000,000 | 251.2     | 200   | 0.082 | 0.970                 | 0.012 | $5.1 \times 10^{-2}$ ( $5.2 \times 10^{-2}$ ) |
| 0.75    | 10,000    | 10.0      | 1,000 | 0.464 | 0.887                 | 0.010 | $8.1 \times 10^{-3}$ ( $8.2 \times 10^{-3}$ ) |
| 0.75    | 100,000   | 17.8      | 500   | 0.339 | 0.924                 | 0.012 | $2.6 \times 10^{-3}$ ( $2.6 \times 10^{-3}$ ) |
| 0.75    | 1,000,000 | 31.6      | 200   | 0.247 | 0.930                 | 0.018 | $8.2 \times 10^{-4}$ ( $8.2 \times 10^{-4}$ ) |
| 0.60    | 1,000,000 | 251.2     | 5,000 | 0.090 | 0.946                 | 0.003 | —                                             |

Table 2: Boundary inference under common temperature  $\tau_n = n^{-\zeta}$ . Coverage is of  $\beta_0 = c(0)$ . Loss is the Monte Carlo mean with the Theorem 1(v) constant in parentheses. The last row is a separate 5,000-replication run.

increasing scale can replace exploration. If the question is the population average effect of a strategy, the logs are informative only to the extent the deployer has paid exploration loss away from the boundary, and increasing scale cannot replace it. Reporting an effect estimated from adaptive logs therefore requires saying which of the two quantities it is.

For design, the relevant choice is not whether to use temperature but how. With one temperature shared by all contexts, the exploration probability decays exponentially in the policy’s confidence, so reaching confident contexts forces a temperature that is costly near the boundary; uniform mixing, context-dependent temperatures, or temperature mixtures with a vanishing weight on a moderate temperature avoid this. Among designs that avoid it, allocating by the value gap saves further loss when gaps are heterogeneous and reasonably well specified (Lemma 1); in our example the saving is about ten percent.

**Limitations.** The policy is known and assignments are logged; the setting is a single decision per unit with independent contexts; the boundary results are one-dimensional and pointwise; the population impossibility excludes models in which boundary data identify the population effect; and the treatment is strategy assignment under a fixed generator, not the realized text. Extending the separation to multi-turn conversations requires defining the reference policy for later turns and the induced distribution of histories, which we leave open. Estimated policies, multidimensional scores and effects of realized text features are also outside the formal results.

## Data availability statement

No new data were collected or analysed. All results are theoretical or based on simulations. The simulation code, the result files (with arguments, random seeds and code commits), plain-text versions of the proofs, and the scripts that generate every number, table and figure in the paper are available in the repository

<https://github.com/kunhailP/when-treatment-talks-back>

(branch `jci-separation`, tagged submission version **[TO CONFIRM: tag or commit]**).

## Author contributions

The author confirms sole responsibility for the conception of the study, the theoretical results and proofs, the simulations, and the preparation of the manuscript, and has approved the final version.

## Conflict of interest

**[TO CONFIRM]** The author states no conflict of interest.

## Funding

**[TO CONFIRM]** This research received no external funding.

## Use of AI-assisted tools

**[TO CONFIRM against the publisher’s policy]** Generative AI tools were used to assist in drafting text, checking proofs, and writing simulation code. The author reviewed all content and takes full responsibility for it.

## References

- [1] Salvi F, Horta Ribeiro M, Gallotti R, West R. On the conversational persuasiveness of GPT-4. *Nature Human Behaviour*. 2025.
- [2] Hackenburg K, Tappin BM, Hewitt L, Saunders E, Black S, Lin H, et al. The levers of political persuasion with conversational artificial intelligence. *Science*. 2025;390(6777):eaea3884. Available from: <https://www.science.org/doi/10.1126/science.aea3884>.
- [3] Narita Y, Yata K. Algorithm as Experiment: Machine Learning, Market Design, and Policy Eligibility Rules; 2021. ArXiv:2104.12909.
- [4] Eckles D, Ignatiadis N, Wager S, Wu H. Noise-Induced Randomization in Regression Discontinuity Designs. *Biometrika*. 2025;112(2):asaf003.
- [5] Li F, Morgan KL, Zaslavsky AM. Balancing Covariates via Propensity Score Weighting. *Journal of the American Statistical Association*. 2018;113(521):390-400.

- [6] Simchi-Levi D, Wang C. Multi-Armed Bandit Experimental Design: Online Decision-Making and Adaptive Inference. In: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics. vol. 206 of PMLR; 2023. .
- [7] Wan R, Kveton B, Song R. Safe Exploration for Efficient Policy Evaluation and Comparison. In: Proceedings of the 39th International Conference on Machine Learning. vol. 162 of PMLR; 2022. .
- [8] Douglas C, Persson J, Provost F. Logging Policy Design for Off-Policy Evaluation; 2026. ArXiv:2605.15108.
- [9] Koppel A, Bhatt S, Zeng S, Ganesh S. The Explore-Exploit Tradeoff Redefined: Balancing Regret and Treatment Effects in Contextual Bandits; 2025. OpenReview K7n5igM0mS. NeurIPS 2025 Workshop on Constrained Optimization for Machine Learning.
- [10] Cochran WG. Sampling Techniques. 3rd ed. New York: Wiley; 1977.
- [11] Cesa-Bianchi N, Gentile C, Lugosi G, Neu G. Boltzmann Exploration Done Right. In: Advances in Neural Information Processing Systems. vol. 30; 2017. .
- [12] Li L, Munos R, Szepesvári C. Toward Minimax Off-Policy Value Estimation. In: Proceedings of the 18th International Conference on Artificial Intelligence and Statistics. vol. 38 of PMLR; 2015. .
- [13] Wang YX, Agarwal A, Dudík M. Optimal and Adaptive Off-Policy Evaluation in Contextual Bandits. In: Proceedings of the 34th International Conference on Machine Learning. vol. 70 of PMLR; 2017. .
- [14] Mou W, Ding P, Wainwright MJ, Bartlett PL. Kernel-Based Off-Policy Estimation without Overlap: Instance Optimality beyond Semiparametric Efficiency; 2023. ArXiv:2301.06240.
- [15] Khan S, Tamer E. Irregular Identification, Support Conditions, and Inverse Weight Estimation. *Econometrica*. 2010;78(6):2021-42.
- [16] Hong H, Leung MP, Li J. Inference on Finite-Population Treatment Effects under Limited Overlap. *The Econometrics Journal*. 2020;23(1):32-47.
- [17] Ma X, Wang J. Robust Inference Using Inverse Probability Weighting. *Journal of the American Statistical Association*. 2020;115(532):1851-60.
- [18] Khan S, Saveski M, Ugander J. Off-Policy Evaluation beyond Overlap: Partial Identification through Smoothness; 2024. ArXiv:2305.11812.
- [19] Athey S, Imbens G, Ji Z. Estimating Causal Effects from Data Generated by Stochastic Algorithms; 2026. ArXiv:2607.05792.

- [20] Fresh A, Shin A. What is the Causal Effect of a Conversation? Estimands and Inference in AI Mediated Conversations; 2026. ArXiv:2607.03597.
- [21] Imai K, Nakamura K. Causal inference with generative artificial intelligence: Application to texts as treatments. Journal of the American Statistical Association. 2026. ArXiv:2410.00903.
- [22] Nakamura K, Imai K. GenAI Powered Dynamic Causal Inference with Unstructured Data; 2026. ArXiv:2605.07834.
- [23] Díaz Muñoz I, van der Laan MJ. Population Intervention Causal Effects Based on Stochastic Interventions. Biometrics. 2012;68(2):541-9.
- [24] Young JG, Hernán MA, Robins JM. Identification, Estimation and Approximation of Risk under Interventions that Depend on the Natural Value of Treatment Using Observational Data. Epidemiologic Methods. 2014;3(1):1-19.
- [25] Gill RD, Levit BY. Applications of the van Trees Inequality: A Bayesian Cramér–Rao Bound. Bernoulli. 1995;1(1–2):59-79.

## A Proof of Theorem 1

Write  $e = e_\tau(H) = \text{sig}(H/\tau) = \mathbb{P}(A = 1 \mid H)$  and  $\omega = e(1 - e) = K(H/\tau)$  with  $K(u) = \exp(u)/\{1 + \exp(u)\}^2$ , and  $m_\tau = \mathbb{E}\omega$ . We use

$$\begin{aligned} \int K &= 1, & \int |u|K(u) du &= 2 \log 2, & \int K(u)\text{sig}(u) du &= \frac{1}{2}, \\ K(u) &\leq \exp(-|u|), & \int_0^\infty v \text{sig}(-v) dv &= \frac{\pi^2}{12}. \end{aligned}$$

**Step 1 (kernel integrals).** For  $\varrho$  bounded and continuous at 0,  $j \geq 1$  and  $r, s \geq 0$ , substituting  $h = \tau u$  and using dominated convergence with dominating function  $\|\varrho\|_\infty \bar{f} K(u)$ ,

$$\mathbb{E}[K(H/\tau)^j (1 - e)^r e^s \varrho(H)] = \tau f(0) \varrho(0) \int K^j \text{sig}(-u)^r \text{sig}(u)^s du + o(\tau).$$

In particular  $m_\tau = \tau f(0)(1 + o(1))$ .

**Step 2 (bias).**  $\beta_{\text{ov},\tau} - \beta_0 = \mathbb{E}[\omega\{c(H) - c(0)\}]/m_\tau$ . On  $|H| \leq u_0$ ,  $|c(h) - c(0)| \leq 2L|h|$  and  $\mathbb{E}[\omega|H|] \leq \bar{f}\tau^2 \int |u|K = 2 \log 2 \bar{f}\tau^2$ . On  $|H| > u_0$ ,  $\omega \leq \exp(-u_0/\tau)$  and  $|c(h) - c(0)| \leq 4B$ . Hence  $|\beta_{\text{ov},\tau} - \beta_0| \leq (4 \log 2 L \bar{f}\tau^2 + 4B \exp(-u_0/\tau))/m_\tau = (4 \log 2 L \bar{f}/f(0))\tau(1 + o(1))$ .

**Step 3 (linearization).** Let  $\bar{\mu}_a = \mathbb{E}[\omega\mu_a]/m_\tau$ , so  $\bar{\mu}_1 - \bar{\mu}_0 = \beta_{\text{ov},\tau}$ . Since  $\mathbb{E}[w_1 | H] = \omega$  and  $\mathbb{E}[w_1 Y | H] = \omega\mu_1$ ,  $\mathbb{E}[w_1(Y - \bar{\mu}_1)] = 0$ , and likewise for arm 0. Define  $T_a = n^{-1} \sum_i w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$  and  $\xi_i = \{w_{1i}(Y_i - \bar{\mu}_1) - w_{0i}(Y_i - \bar{\mu}_0)\}/m_\tau$ . Then  $\hat{m}_a - \bar{\mu}_a = (m_\tau/\hat{D}_a)T_a$  and  $\mathbb{E}\xi = 0$ . Because  $w_1 w_0 = 0$  and  $\mathbb{E}[A(1-e)^2 | H] = \omega(1-e)$ ,

$$s_n^2 := \text{Var}\xi = m_\tau^{-2} \mathbb{E}[\omega\{(1-e)(\sigma_1^2 + (\mu_1 - \bar{\mu}_1)^2) + e(\sigma_0^2 + (\mu_0 - \bar{\mu}_0)^2)\}].$$

By Step 1,  $\bar{\mu}_a \rightarrow \mu_a(0)$ ; the terms in  $(\mu_a - \bar{\mu}_a)^2$  are  $o(\tau)$  by dominated convergence; and the variance terms equal  $\tau f(0)\sigma_a^2(0)/2 + o(\tau)$ . Hence  $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau f(0)\}(1 + o(1))$ .

**Step 4 (central limit theorem).** Since  $0 \leq w_a \leq 1$ ,  $w_a^4 \leq w_a$ , and  $\mathbb{E}[(Y - \bar{\mu}_a)^4 | A, H] \leq C(M, B)$ , we get  $\mathbb{E}[w_a^4(Y - \bar{\mu}_a)^4] \leq C m_\tau$  and  $\mathbb{E}\xi^4 \leq C' m_\tau^{-3}$ . The Lyapunov ratio  $n^{-1}\mathbb{E}\xi^4/s_n^4 \asymp (n\tau)^{-1} \rightarrow 0$ , so  $n^{-1/2} \sum \xi_i/s_n \Rightarrow N(0, 1)$ . Also  $\text{Var}(\hat{D}_a)/m_\tau^2 \leq 1/(nm_\tau) \rightarrow 0$ , so  $\hat{D}_a/m_\tau \rightarrow 1$  in probability and  $m_\tau/\hat{D}_a - 1 = O_p((n\tau)^{-1/2})$ . Now

$$\hat{\beta} - \beta_{\text{ov},\tau} = (T_1 - T_0) + (m_\tau/\hat{D}_1 - 1)T_1 - (m_\tau/\hat{D}_0 - 1)T_0,$$

with  $T_1 - T_0 = n^{-1} \sum \xi_i$  and  $T_a = O_p((n\tau)^{-1/2})$ . The remainder is  $O_p((n\tau)^{-1}) = o_p(s_n/\sqrt{n})$ , and Slutsky's lemma gives (i).

**Step 5 (variance estimator).** Write  $\xi_{ai} = w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$ . First,  $n^{-1} \sum \xi_i^2/s_n^2$  has mean one and variance at most  $\mathbb{E}\xi^4/(n s_n^4) \asymp (n\tau)^{-1}$ . Second,

$$\hat{\psi}_{1i} - \xi_{1i} = w_{1i}(Y_i - \bar{\mu}_1)(1/\hat{D}_1 - 1/m_\tau) + w_{1i}(\bar{\mu}_1 - \hat{m}_1)/\hat{D}_1.$$

The mean square of the first term is  $(n^{-1} \sum \xi_{1i}^2)(m_\tau/\hat{D}_1 - 1)^2 = O_p(s_n^2)O_p((n\tau)^{-1})$ . For the second,  $w_1^2 \leq w_1$  gives mean square at most  $(\hat{m}_1 - \bar{\mu}_1)^2/\hat{D}_1$ ; by Step 4 and  $\hat{D}_1 = \tau f(0)(1 + o_p(1))$  this is  $O_p(1/(n\tau^2))$ , which is  $O_p((n\tau)^{-1})$  relative to  $s_n^2 \asymp \tau^{-1}$ . Arm 0 is identical. By Minkowski's and the Cauchy-Schwarz inequalities,  $|n^{-1} \sum \hat{\psi}_i^2 - n^{-1} \sum \xi_i^2| = o_p(s_n^2)$ , which gives (iii).

**Step 6 (centering at  $\beta_0$ ).**  $(\beta_{\text{ov},\tau} - \beta_0)/(s_n/\sqrt{n}) = O(\tau)O((n\tau)^{1/2}) = O((n\tau^3)^{1/2})$ . With (i), (iii) and Slutsky's lemma this gives (iv).

**Step 7 (exploration loss).** The off-greedy probability is  $q_\tau(h) = \text{sig}(-|h|/\tau)$ , which equals  $e_\tau(h)$  for  $h < 0$  and  $1 - e_\tau(h)$  for  $h > 0$ . On  $|h| \leq u_0$ ,  $n\mathbb{E}[\phi(|H|)\text{sig}(-|H|/\tau)] \leq n\bar{\kappa}\bar{f} \cdot 2\tau^2 \int_0^\infty v \text{sig}(-v)dv = n\bar{\kappa}\bar{f}\tau^2\pi^2/6$ , and the rest is at most  $n\bar{G}\exp(-u_0/\tau)$ . In the strong form, on  $v \leq u_0/\tau$ ,  $\phi(\tau v)/\tau \rightarrow \kappa v$  and  $f(\tau v) \rightarrow f(0)$  with dominating function  $\bar{\kappa}v\bar{f}\text{sig}(-v)$ , which gives the asymptotic equality. For  $\tau_n = n^{-\zeta}$ :  $n\tau_n \rightarrow \infty$  iff  $\zeta < 1$ ;  $n\tau_n^3 \rightarrow 0$  iff  $\zeta > 1/3$ ;  $n\tau_n^2 \rightarrow 0$  iff  $\zeta > 1/2$ .  $\square$

**Smoother effects.** If  $|c(h) - c(0) - c'(0)h| \leq M_2 h^2$  and  $f$  is  $L_f$ -Lipschitz near 0, then  $\mathbb{E}[\omega H] = \tau^2 \int u K(u)\{f(\tau u) - f(0)\}du = O(\tau^3)$  up to an exponentially small tail, and the quadratic remainder contributes at most  $M_2 \bar{f} \tau^3 \int u^2 K/m_\tau$ ; the bias is  $O(\tau^2)$ . Differentiability of  $c$  alone is not enough:  $c(h) = 1 + |h|^{3/2}$  gives bias  $\tau^{3/2} \int |u|^{3/2} K(u)du (1 + o(1))$ .

## B Proofs for Section 4

**Proof of Theorem 2.** *Submodel.* For  $\vartheta \in [-B, B]^K$  set  $\mu_{a^*(h)}^\vartheta(h) = 0$ , and on  $S_k$  set  $\mu_{\bar{a}(h)}^\vartheta(h) = s_k \vartheta_k$  with  $s_k = +1$  if  $\bar{a} = 1$  on  $S_k$  and  $s_k = -1$  otherwise; elsewhere  $\mu_{\bar{a}(h)}^\vartheta(h) = 0$ . Then  $\mu^\vartheta \in \mathcal{M}(B, \sigma)$  and  $\theta(\mu^\vartheta) = \sum_k f_k \vartheta_k$ .

*Information.* The sample density is  $\prod_i dF(H_i) \kappa_i(A_i \mid H_i, \mathcal{F}_{i-1}) \varphi_\sigma(Y_i - \mu_{A_i}^\vartheta(H_i))$ , where the assignment kernels  $\kappa_i$  are known functions of observed data. The score for  $\vartheta_k$  is  $\sum_i \mathbf{1}\{H_i \in S_k, A_i = \bar{a}(H_i)\} s_k (Y_i - s_k \vartheta_k) / \sigma^2$ . Its summands are martingale differences, so the Fisher information matrix is diagonal with  $I_k(\vartheta) = \mathbb{E}_\vartheta N_k / \sigma^2$ , where  $N_k = \sum_i \mathbf{1}\{H_i \in S_k, A_i = \bar{a}(H_i)\}$ .

*van Trees.* Take the product prior with marginal density  $B^{-1} \cos^2(\pi x / (2B))$  on  $[-B, B]$ , whose Fisher information is  $J = \pi^2 / B^2$ . With  $\bar{N}_k = \int \mathbb{E}_\vartheta N_k \lambda(d\vartheta)$ , the multivariate van Trees inequality [25], maximized over the direction vector, gives

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \sum_k \frac{f_k^2}{\bar{N}_k / \sigma^2 + J} = \sum_k \frac{f_k \sigma^2}{a_k}, \quad a_k = \frac{\bar{N}_k + \sigma^2 J}{f_k}.$$

*Cauchy-Schwarz and loss.* Writing  $\sigma \sum_k f_k \sqrt{g_k} = \sum_k \sqrt{f_k g_k a_k} \cdot \sigma \sqrt{f_k / a_k}$ ,

$$\sigma^2 \left( \sum_k f_k \sqrt{g_k} \right)^2 \leq \left( \sum_k f_k g_k a_k \right) \left( \sum_k f_k \sigma^2 / a_k \right).$$

Here  $\sum_k f_k g_k a_k = \sum_k g_k \bar{N}_k + \sigma^2 J \sum_k g_k$ , and since  $g \geq g_k$  on  $S_k$ ,  $\sum_k g_k N_k \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$ , so  $\sum_k g_k \bar{N}_k \leq \sup R_n$ . This proves (1).

*Consequences.* With  $K = 1$  and  $S_1 = \{g \geq \gamma\} \cap \{a^* = j\}$  of positive mass, (1) gives  $\sup R_n \geq g_1 \{\sigma^2 f_1^2 / \sup \mathbb{E}(\hat{\theta} - \theta)^2 - \sigma^2 \pi^2 / B^2\} \rightarrow \infty$ . For the constant, fix a partition into level sets of  $g$  intersected with the two greedy sides; (1) gives  $\liminf \delta_n^2 \sup R_n \geq \sigma^2 (\sum f_k \sqrt{g_k})^2$ , and lower sums  $\sum F(S) \inf_S \sqrt{g}$  increase to  $\mathbb{E} \sqrt{g}$  as the level mesh shrinks.  $\square$

**Proof of Proposition 1.** The Horvitz-Thompson estimator is unbiased with variance at most  $n^{-1}(\sigma^2 + B^2) \mathbb{E}[1/e + 1/(1-e)] = n^{-1}(\sigma^2 + B^2) \mathbb{E}[1/q + 1/(1-q)] \leq n^{-1}(\sigma^2 + B^2)(\mathbb{E}[1/q] + 2)$ , since  $\{e, 1-e\} = \{q, 1-q\}$  and  $q \leq 1/2$ . Pointwise  $1/q = \max\{2, \sqrt{g}/c_n\} \leq \sqrt{g}/c_n + 2$ , and the choice of  $c_n$  gives  $\mathbb{E}[1/q] + 2 \leq n \delta_n^2 / (\sigma^2 + B^2)$ . The loss is  $n \mathbb{E}[gq] \leq n c_n \mathbb{E} \sqrt{g}$ . For uniform mixing with  $C = \sigma^2 + B^2$  and  $q = C / (n \delta_n^2 - 2C) \leq 1/2$ , the same variance bound gives  $C(1/q + 2)/n = \delta_n^2$ , and the loss is  $nq \mathbb{E}[g] = C \mathbb{E}[g] n / (n \delta_n^2 - 2C)$ . Without the correction,  $q = C / (n \delta_n^2)$  gives risk  $C / \{nq(1-q)\} > \delta_n^2$  at  $\mu_0 = \mu_1 \equiv B$ , where the bound is attained.  $\square$

## C Proof of Theorem 3

The function  $b$  satisfies  $0 \leq b \leq 1$ , is  $\pi/(u_2 - u_1)$ -Lipschitz, vanishes on  $(-\infty, u_1]$ , and has  $\mathbb{E}b(H) > 0$  because  $H$  has a density and  $F([u_1, u_2]) > 0$ . Hence  $|tb| \leq B$  and  $tb$

is  $L$ -Lipschitz for  $|t| \leq t_{\max}$ , so  $\mu^t \in \mathcal{M}_D$ ;  $c^t = tb$ ,  $\beta_0(\mu^t) = 0$ , and  $\theta(\mu^t) = t\mathbb{E}b(H)$ . Every perturbed observation is off-greedy, since  $a^* = 1$  on  $(0, \infty)$ , with  $H \in [u_1, u_2]$ . Let  $N = \sum_i \mathbf{1}\{A_i = 0, H_i \in [u_1, u_2]\}$ .

(S1) is Theorem 1, because every member of  $\mathcal{M}_D$  satisfies Assumption 1.

(S2). The sample densities under  $\mu^0$  and  $\mu^t$  coincide on  $E = \{N = 0\}$ , so  $P_0(E) = P_t(E)$ , and for any event  $A$ ,  $|P_0(A) - P_t(A)| = |P_0(A \cap E^c) - P_t(A \cap E^c)| \leq \max\{P_0(E^c), P_t(E^c)\} = P_0(E^c)$ . Thus  $\text{TV} \leq P_0(N \geq 1) \leq \mathbb{E}_0 N$ . Because  $g \geq \gamma$   $F$ -almost surely on  $[u_1, u_2]$  and  $H_i \sim F$ ,  $\gamma N \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$  almost surely, so  $\mathbb{E}_0 N \leq R_n(\mu^0)/\gamma$ . Le Cam's two-point argument with separation  $t_1 \mathbb{E}b$  gives the displayed bound.

(S3). The score of  $t \mapsto \mu^t$  is  $\sum_i \mathbf{1}\{A_i = 0\}(-b(H_i))(Y_i + tb(H_i))/\sigma^2$ , with information  $I(t) = \mathbb{E}_t \sum_i \mathbf{1}\{A_i = 0\} b(H_i)^2 / \sigma^2 \leq \mathbb{E}_t N / \sigma^2 \leq R_n(\mu^t) / (\gamma \sigma^2)$ . With the prior  $t_{\max}^{-1} \cos^2(\pi t / (2t_{\max}))$  on  $[-t_{\max}, t_{\max}]$ , whose information is  $\pi^2 / t_{\max}^2$ , the van Trees inequality for  $\psi(t) = t\mathbb{E}b$  gives the bound.  $\square$

## D Proofs for Section 6

**Proof of Theorem 4.** Take the single cell  $S = \{|\Delta| \geq G - \eta', a^* = j\}$ , with  $j$  attaining  $\nu = \nu_{\eta'}$ . There  $q_\tau \leq \exp(-(G - \eta')/\tau)$ , so  $\bar{N} \leq n\nu \exp(-(G - \eta')/\tau)$ , and the van Trees step of Theorem 2 with  $K = 1$  gives  $\delta_n^2 \geq \nu^2 \sigma^2 / (n\nu \exp(-(G - \eta')/\tau) + \sigma^2 \pi^2 / B^2)$ . Once  $\delta_n^2 \leq \nu^2 B^2 / (2\pi^2)$ , this forces  $n\nu \exp(-(G - \eta')/\tau) \geq \nu^2 \sigma^2 / (2\delta_n^2)$ , that is,  $\tau_n \geq (G - \eta') / \log\{2n\delta_n^2 / (\nu \sigma^2)\}$ , where the logarithm is positive because  $2n\delta_n^2 > \nu \sigma^2$ . Nothing here requires  $\delta_n \rightarrow 0$ . On the other hand

$$R_n = n\mathbb{E}[\phi(|\Delta|)\text{sig}(-|\Delta|/\tau)] \geq n\kappa f_{\min} \tau^2 \int_0^{u_0/\tau} v \text{sig}(-v) dv \geq n\kappa f_{\min} c_0 \min\{\tau, u_0\}^2.$$

Combining the two bounds proves the claim. With finitely many score values, the same argument gives  $\tau \geq \Delta_{\max} / \log(Cn\delta^2)$  for  $\delta$  small enough. The  $\Delta_{\min}$  cell, with mass  $f$ , then contributes at least

$$nf\phi(\Delta_{\min}) \exp(-\Delta_{\min}/\tau)/2. \quad \square$$

**Remark 1.** Write  $u = |H|$ ,  $r(u) = \text{sig}(-u)$  for the off-greedy probability at temperature 1, and  $q_n = w_n r + (1 - w_n) \text{sig}(-nu)$  for the mixture. Since  $q_n \geq w_n r$ ,  $\mathbb{E}[1/q_n]/n \leq \delta^2$  for  $w_n = \mathbb{E}[1/r]/(n\delta^2)$ , and  $n\mathbb{E}[uq_n] \leq \mathbb{E}[1/r]\mathbb{E}[ur]/\delta^2 + O(1/n)$ . With  $w_n = \mathbb{E}[1/r]/\{n\delta_n^2/(\sigma^2 + B^2) - 2\}$  the Horvitz–Thompson risk is at most  $\delta_n^2$  uniformly over  $\mathcal{M}(B, \sigma)$ .

**Proof of Proposition 2.** Solving  $\text{sig}(-|\Delta|/\tau) = q$  gives the formula. If  $g(h) = 0$ , then  $q^*(h) = 1/2$ , which is implementable by  $\tau = \infty$  (and by any  $\tau$  when  $\Delta(h) = 0$ ). If  $\Delta(h) \neq 0$  and  $g(h) > 0$ , then  $q^*(h) = c_n / \sqrt{g(h)} < 1/2$  once  $c_n < \sqrt{g(h)}/2$ , and  $\log\{(1 - q^*)/q^*\} = \log(1/c_n) + \frac{1}{2} \log g(h) + \log(1 - q^*)$  with  $c_n \rightarrow 0$ , which gives the limit.  $\square$

**Proof of Lemma 1.** With the cap slack, let  $q = c/\sqrt{\hat{g}}$  and  $\mathbb{E}[1/q] = T$ . The loss is  $n\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/T$ , against  $n(\mathbb{E}\sqrt{g})^2/T$  for  $\hat{g} = g$ . With  $s = (\hat{g}/g)^{1/2} \in [\sqrt{a}, \sqrt{b}]$  and the probability measure  $w \propto \sqrt{g} dF$ , the ratio is  $\mathbb{E}_w[s]\mathbb{E}_w[1/s]$ , which Kantorovich's inequality bounds by  $(\sqrt{a} + \sqrt{b})^2/(4\sqrt{ab}) = K(b/a)$ . If  $w$  has a set of mass one half, setting  $s = \sqrt{a}$  there and  $s = \sqrt{b}$  elsewhere attains the bound. With atoms the bound need not be attained: for  $F = (0.99, 0.01)$ ,  $g = (1, 100)$  and  $\rho = 36$ , uniform mixing costs 1.675 times the optimum, below  $K(36) = 2.042$ , yet every admissible  $\hat{g}$  costs at most 1.347 times the optimum.  $\square$